Squared partial traces and the nuclear norm: a multipartite inequality and the first-order Werner bound

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Abstract

We prove that every operator Z on a finite-dimensional tensor product $H_1 \otimes \cdots \otimes H_n$ satisfies

$$\sum_{i=1}^{n} \|\mathrm{Tr}_i\, Z\|_2^2 \;\le\; (n-1)\|Z\|_2^2 + \|Z\|_1^2,$$

where Tr_i is the partial trace over the ith factor. The inequality is sharp. It needs no positivity and no rank assumption. For $n = 2$ it follows from a recent bound of Costa Rico and Wolf; for $n \ge 3$ it is new.

For operators of rank at most two it gives $\sum_i \|\mathrm{Tr}_i\, C\|_2^2 \le (n+1)\|C\|_2^2$. This is Assumption (i) of Chattopadhyay and Sarkar from 2006, i.e. the first-order part of the n-copy distillability problem for Werner states at the endpoint $\alpha = 1/2$. Iterating the inequality gives all of their assumed bounds for partial traces over m sites. For density matrices it is a Han-type inequality for the linear (Tsallis-2) entropy, $\sum_i S_L(\rho_i) \ge (n-1)S_L(\rho)$, which holds even though strong subadditivity fails for this entropy.

An induction on n reduces the inequality to a single statement about block operators. We prove that statement with a balanced factorisation and second compound matrices. The method does not reach the endpoint itself: every rank-free relaxation of the endpoint inequality we tried fails already for two or three copies. In the remarks we split the multiplicative ("no collective gain") form of the problem into n inequalities, of which ours is the first and the endpoint the last. For three copies the other two follow from the two-copy theorem and Tomar's computer-assisted three-copy theorem. So, given that theorem, the multiplicative form holds for three copies. We also prove that the two-copy inequality has a margin linear in the thickness of the co-range, which answers a question of Tomar, and that for normal operators it holds with the sharp constant $1 - 1/\sqrt{2}$, better than the $\frac{1}{4}$ Tomar conjectured.

1 Introduction

Let F be the swap on $\mathbb{C}^d \otimes \mathbb{C}^d$. For $\alpha \in [-1, 1]$ let $\rho_\alpha \propto I - \alpha F$ be the Werner state. It has a non-positive partial transpose (NPT) for $\alpha > 1/d$ and is one-copy distillable for $\alpha > 1/2$. Whether the states with $1/d < \alpha \le 1/2$ can be distilled from some number of copies is the NPT bound-entanglement problem [1, 2, 3].

The two-copy case was settled in 2026 by several groups [4, 5, 6, 7]. Three copies have since been settled at the endpoint with computer assistance in every dimension [9], and Wu and Zou proved the three-copy endpoint inequality for positive semidefinite and for normal operators of rank two, and for operators whose range and co-range lie in one plane [8]. The general n-copy question remains open.

By [11, 5], $\rho_{1/2}$ is n-copy undistillable if and only if

$$q_n(C) \;=\; \sum_{J\subseteq\{1,\dots,n\}} \Big(-\tfrac{1}{2}\Big)^{|J|} \,\|\mathrm{Tr}_J\, C\|_2^2 \;\geq\; 0 \tag{1}$$

for every operator C on $(\mathbb{C}^d)^{\otimes n}$ of rank at most two. Here Tr_J is the partial trace over the factors in J.

Long before these results, Chattopadhyay and Sarkar [12] proposed bounds for Schmidt-rank-two vectors ψ in $(\mathbb{C}^d\otimes\mathbb{C}^d)^{\otimes n}$. Let ψ be the vectorisation of C, normalised so that $\|C\|_2 = 1$. With the unnormalised projector $P^+ = \sum_{j,k} |jj\rangle\langle kk|$ (trace d) one has $\langle\psi|P_J^+\otimes I|\psi\rangle = \|\mathrm{Tr}_J\, C\|_2^2$, where P_J^+ acts on the pairs in J. Their bounds then read

$$S_m(C) := \sum_{|J|=m} \|\mathrm{Tr}_J\, C\|_2^2 \;\leq\; \big[\tbinom{n-1}{m} + 2\tbinom{n-1}{m-1}\big]\,\|C\|_2^2, \qquad m = 1,\dots,n, \tag{2}$$

for every C of rank at most two.

Chattopadhyay and Sarkar presented these bounds as assumptions supported by numerics. Earlier versions of their paper claimed more; Pankowski, Piani and the Horodeckis later noted that the proof had gaps [13]. Two cases are now known.

- The case $n = 2$ is $\|\mathrm{Tr}_1\, C\|_2^2 + \|\mathrm{Tr}_2\, C\|_2^2 \leq 3\|C\|_2^2$ [16, Eq. (32)]. It also follows from the sharper rank-two inequalities of [6, 7].
- The case $m = n-1$ involves only one-site marginals. It follows from Corollary 2 of Costa Rico and Wolf [16]: $\sum_i \|\mathrm{Tr}_{\hat{\imath}}\, M\|_2^2 \leq (1 + r(n-1))\|M\|_2^2$ for operators M of rank r.

Their separate two-copy assumption (Eq. (3) of [12]) follows from the rank-two bipartite bounds of [6, 7].

The case $m = 1$, $\sum_i \|\mathrm{Tr}_i\, C\|_2^2 \leq (n+1)\|C\|_2^2$, is the first-order part of (1): it is the derivative at $t = 0$ of the conjectured identity $\min_C \sum_J (-t)^{|J|}\|\mathrm{Tr}_J\, C\|_2^2 = (1-2t)(1-t)^{n-1}\|C\|_2^2$ for rank-two C.

Our main result proves all of (2), in particular the first-order case $m = 1$ for every $n \geq 3$, as a consequence of one inequality that holds for every operator.

Section 5 contains further results; we mention three. The conjectured identity above splits into n inequalities, of which ours is the first and (1) the last. For three copies the middle one follows from the two-copy theorem and Tomar's three-copy theorem. Finally, for two copies we prove that $q_2(C)$ is bounded below by a constant times the "thickness" of the co-range of C (Proposition 7), which answers a question of Tomar [10]; when the range and co-range of C lie in one plane, for instance for normal C, we obtain the sharp constant $1 - 1/\sqrt{2}$, which beats the $\frac{1}{4}$ he conjectured. Tomar proved this constant on a different class of planes [10, Thm. 7.2].

Partial-trace norm inequalities have a long history.

- Audenaert proved subadditivity of q-entropies for $q > 1$ [14].
- Rastegin studied unitarily invariant norms of partial traces [15].
- Costa Rico and Wolf extended Audenaert's inequality to general matrices and several tensor factors [16]. Their bounds, such as $\sum_i \|M_i\|_p \leq (n-1)\|M\|_1 + \|M\|_p$ (their Prop. 9), concern the one-site marginals M_i. In the notation above they control S_{n-1}, whereas the first-order term is S_1. For $n = 2$ the two kinds of marginal coincide. There Prop. 9 with $p = 2$, followed by an elementary squaring step, gives the case $n = 2$ of Theorem 1 below.

- The two-copy proofs rely on sharp bipartite inequalities for operators of bounded rank [6, 7].
- Wu and Zou proved a sharp rank-free bipartite inequality, $\|\mathrm{Tr}_K X\|_2^2 + \|\mathrm{Tr}_L X\|_2^2 \le \|X\|_1^2 + |\mathrm{Tr} X|^2$ [8, Thm. 1.1]. Neither this nor our $n = 2$ case implies the other: the identity favours ours, and traceless operators favour theirs.
- The first step of our block lemma is the second-compound estimate of Wu and Zou [8, Lemma 5.1], in a homogeneous form.

For $n \ge 3$ we are not aware of a previous statement of Theorem 1, of Lemma 6, or of the bounds (2) with $m < n - 1$.

2 Results

Throughout, $H = H_1 \otimes \cdots \otimes H_n$ with $\dim H_i < \infty$. We write $\|\cdot\|_2$ for the Hilbert–Schmidt norm, $\|\cdot\|_1$ for the trace (nuclear) norm and $\langle A, B\rangle = \mathrm{Tr}(A^* B)$.

Theorem 1. *For every operator Z on H,*

$$\sum_{i=1}^{n} \|\mathrm{Tr}_i Z\|_2^2 \;\le\; (n-1)\|Z\|_2^2 + \|Z\|_1^2. \tag{3}$$

Equality holds for $Z = e^{i\theta} P \otimes |c\rangle\langle c|$ with $P \ge 0$ on one factor and c a product unit vector on the others.

Since $\|C\|_1^2 \le \mathrm{rank}(C)\, \|C\|_2^2$, Theorem 1 has an immediate consequence.

Corollary 2. *If $\mathrm{rank}\, C \le r$, then $\sum_i \|\mathrm{Tr}_i C\|_2^2 \le (n + r - 1)\|C\|_2^2$. For $r = 2$ this is (2) with $m = 1$, for every n and every local dimension, including nonnormal C.*

The bound is sharp for every r: take $C = I_U \otimes |c\rangle\langle c|$ with U an r-dimensional subspace of one factor and c a product vector. It is the first-order term of the natural rank-r form of the multiplicativity conjecture, $\min\{\sum_J (-t)^{|J|} \|\mathrm{Tr}_J C\|_2^2 : \mathrm{rank}\, C \le r,\ \|C\|_2 = 1\} = (1 - rt)(1 - t)^{n-1}$, whose derivative at $t = 0$ is $-(n + r - 1)$. The case $r = 2$, $t = 1/2$ is (1), i.e. the statement that the map $X \mapsto \mathrm{Tr}(X) I - X/2$ is tensor-stably 2-positive.

In the language of Schmidt-number witnesses [16], Corollary 2 says the following. On $(\mathbb{C}^d \otimes \mathbb{C}^d)^{\otimes n}$ the operator

$$W_r = (n + r - 1)\, I - \sum_{i=1}^{n} P^{+}_{A_i B_i} \otimes I_{\mathrm{rest}}$$

is nonnegative on every vector of Schmidt rank at most r across $A|B$. For $r < d$ it is negative on a vector of Schmidt rank $r + 1$. Equivalently, $X \mapsto (n + r - 1)\,\mathrm{Tr}(X)\, I - \sum_i \mathrm{Tr}_{\hat{\imath}}(X) \otimes I_{\hat{\imath}}$ is r-positive but not $(r + 1)$-positive. For $r = 2$ this is the operator of [12]. Costa Rico and Wolf give the analogous family in which only one copy is left untraced [16].

Corollary 3. *For every operator Z on H and every $0 \le m \le n$,*

$$S_m(Z) \le \tbinom{n-1}{m} \|Z\|_2^2 + \tbinom{n-1}{m-1} \|Z\|_1^2,$$

with $\binom{n-1}{-1} = 0$. Equality holds for $Z = P \otimes |c\rangle\langle c|$ as in Theorem 1. In particular all the bounds (2) hold.

Corollary 4. *For every density matrix ρ on H, with $\rho_{\hat{\imath}} = \operatorname{Tr}_i \rho$ and $S_L(\sigma) = 1 - \operatorname{Tr}\sigma^2$,*

$$\sum_{i=1}^{n} \operatorname{Tr}\rho_{\hat{\imath}}^2 \le (n-1)\operatorname{Tr}\rho^2 + 1, \qquad \textit{equivalently} \qquad \sum_{i=1}^{n} S_L(\rho_{\hat{\imath}}) \ge (n-1) S_L(\rho).$$

For the von Neumann entropy this is Han's inequality. Its usual proof uses strong subadditivity, which fails for S_L. The case $n = 2$ is subadditivity of S_L [14]. For diagonal ρ, Corollary 4 is the case $q = 2$ of the classical Han inequality for Tsallis entropies [17].

Corollary 5. *Let A_0 be an auxiliary system. For states σ on $A_0 \otimes H$ write σ_0 and σ_{0i} for the marginals on A_0 and on $A_0 \otimes H_i$. The map*

$$\Lambda(\sigma) = \operatorname{Tr}(\sigma)\, I + (n-1)\,\sigma_0 \otimes I_H - \sum_{i=1}^{n} \sigma_{0i} \otimes I_{H_i}$$

is positive. Consequently every state on $A_0 \otimes H$ satisfies $\sum_i \operatorname{Tr}\sigma_{0i}^2 \le (n-1)\operatorname{Tr}\sigma_0^2 + 1$.

For $n = 1$, Λ is the reduction map. For $n \ge 2$ one checks numerically that Λ is not decomposable, so its positivity is not a consequence of positivity and partial transposition alone.

3 The block lemma

Lemma 6. *Let K be finite-dimensional and let $Z = \sum_{a,b=1}^{d} |a\rangle\langle b| \otimes Z_{ab}$ be an operator on $\mathbb{C}^d \otimes K$ with blocks Z_{ab} on K. Then*

$$\sum_{a,b} \|Z_{ab}\|_1^2 + \Big\| \sum_a Z_{aa} \Big\|_2^2 \;\le\; \sum_{a,b} \|Z_{ab}\|_2^2 + \|Z\|_1^2. \tag{4}$$

Equality holds for $Z = |v\rangle\langle v| \otimes W$ with any unit v and any W.

Proof. If $Z = 0$ there is nothing to prove, so let $r = \operatorname{rank} Z \ge 1$. Let $Z = U\Sigma V^*$ be a compact singular value decomposition, with $\Sigma > 0$ diagonal of size r. Put $P = U\Sigma^{1/2}$ and $Q = V\Sigma^{1/2}$. Then $Z = PQ^*$ and $P^*P = Q^*Q = \Sigma$.

Let $P_a = (\langle a| \otimes I_K)P$ and $Q_b = (\langle b| \otimes I_K)Q$. These are maps from $\mathbb{C}^r$ to K, and $Z_{ab} = P_a Q_b^*$. Put $R_a = P_a^* P_a$ and $S_b = Q_b^* Q_b$. Then $\sum_a R_a = \sum_b S_b = \Sigma$. Write

$$\alpha_a = \operatorname{Tr} R_a, \quad p_a = \operatorname{Tr} R_a^2, \quad \beta_b = \operatorname{Tr} S_b, \quad q_b = \operatorname{Tr} S_b^2,$$

so that $0 \le p_a \le \alpha_a^2$ and $0 \le q_b \le \beta_b^2$. Then $\|Z\|_1 = \operatorname{Tr}\Sigma = \sum_a \alpha_a = \sum_b \beta_b$, and therefore $\|Z\|_1^2 = \sum_{a,b} \alpha_a \beta_b$.

Step 1 (off-diagonal and diagonal blocks alike).

- For any matrix M with singular values s_j, $\|M\|_1^2 - \|M\|_2^2 = 2\sum_{j<k} s_j s_k = 2\|\wedge^2 M\|_1$. Here $\wedge^2$ is the second compound: for a linear map A, $\wedge^2 A$ is the induced map on antisymmetric tensors. It vanishes when the domain or the range has dimension one.
- The compound is multiplicative and commutes with adjoints, also for rectangular maps (the Cauchy–Binet formula). So $\wedge^2 Z_{ab} = (\wedge^2 P_a)(\wedge^2 Q_b)^*$.
- By Hölder's inequality, $\|XY^*\|_1 \le \|X\|_2 \|Y\|_2$.

- Moreover $\|\wedge^2 P_a\|_2^2 = \operatorname{Tr} \wedge^2(P_a^* P_a) = \frac{1}{2}(\alpha_a^2 - p_a)$, and likewise for Q_b.

Hence

$$\|Z_{ab}\|_1^2 - \|Z_{ab}\|_2^2 \le \sqrt{(\alpha_a^2 - p_a)(\beta_b^2 - q_b)} \le \alpha_a \beta_b - \sqrt{p_a q_b}. \tag{5}$$

The first inequality is [8, Lemma 5.1] in homogeneous form. For the second, both sides are non-negative, and after squaring it reduces to $\alpha_a^2 q_b + \beta_b^2 p_a \ge 2\alpha_a \beta_b \sqrt{p_a q_b}$.

Step 2 (the diagonal blocks). By Cauchy–Schwarz,

$$\|Z_{aa}\|_2^2 = \operatorname{Tr}(Q_a P_a^* P_a Q_a^*) = \operatorname{Tr}(R_a S_a) \le \|R_a\|_2 \|S_a\|_2 = \sqrt{p_a q_a}.$$

By the triangle inequality and Cauchy–Schwarz again,

$$\Big\| \sum_a Z_{aa} \Big\|_2 \le \sum_a (p_a q_a)^{1/4} \le \Big(\sum_a \sqrt{p_a} \Big)^{1/2} \Big(\sum_a \sqrt{q_a} \Big)^{1/2}. \tag{6}$$

Step 3 (conclusion). Let Δ be the right side of (4) minus the left side. Using $\|Z\|_1^2 = \sum_{a,b} \alpha_a \beta_b$, then (5) and (6),

$$\begin{aligned} \Delta &= \sum_{a,b} \big[\alpha_a \beta_b - \big(\|Z_{ab}\|_1^2 - \|Z_{ab}\|_2^2 \big) \big] - \Big\| \sum_a Z_{aa} \Big\|_2^2 \\ &\ge \sum_{a,b} \sqrt{p_a q_b} - \Big(\sum_a \sqrt{p_a} \Big) \Big(\sum_a \sqrt{q_a} \Big) = 0. \end{aligned}$$

□

4 Proofs

Proof of Theorem 1. We use induction on n. For $n = 1$ the claim is $|\operatorname{Tr} Z|^2 \le \|Z\|_1^2$.

For $n \ge 2$ write $Z = \sum_{a,b} |a\rangle\langle b| \otimes Z_{ab}$ with respect to an orthonormal basis of H_n, with blocks Z_{ab} on $H_1 \otimes \cdots \otimes H_{n-1}$. Then:

- $\operatorname{Tr}_i Z = \sum_{a,b} |a\rangle\langle b| \otimes \operatorname{Tr}_i Z_{ab}$ for $i < n$;
- $\operatorname{Tr}_n Z = \sum_a Z_{aa}$;
- $\|Z\|_2^2 = \sum_{a,b} \|Z_{ab}\|_2^2$.

The induction hypothesis, applied to each block, gives

$$\sum_{i<n} \|\operatorname{Tr}_i Z\|_2^2 = \sum_{a,b} \sum_{i<n} \|\operatorname{Tr}_i Z_{ab}\|_2^2 \le (n-2)\|Z\|_2^2 + \sum_{a,b} \|Z_{ab}\|_1^2.$$

Adding $\|\operatorname{Tr}_n Z\|_2^2$ and applying Lemma 6, with $K = H_1 \otimes \cdots \otimes H_{n-1}$ and $d = \dim H_n$, gives (3). Local dimensions may differ, since Lemma 6 allows any d and K.

For the equality case, let $Z = P \otimes |c\rangle\langle c|$ with $P \ge 0$ on the first factor. Then:

- $\|\operatorname{Tr}_1 Z\|_2^2 = (\operatorname{Tr} P)^2 = \|P\|_1^2$;
- $\|\operatorname{Tr}_i Z\|_2^2 = \|P\|_2^2$ for $i \ge 2$;
- $\|Z\|_2 = \|P\|_2$ and $\|Z\|_1 = \|P\|_1$.

So both sides of (3) equal $(n-1)\|P\|_2^2 + \|P\|_1^2$. □

Proof of Corollary 3. We use induction on m, for all n at once. For $m = 0$ the claim is trivial, and for $m = 1$ it is Theorem 1. Let $m \geq 2$, so $n \geq 2$.

Every J with $|J| = m$ satisfies

$$\sum_{i \in J} \|\mathrm{Tr}_{J\setminus i}(\mathrm{Tr}_i\, Z)\|_2^2 = m\|\mathrm{Tr}_J\, Z\|_2^2.$$

Hence $mS_m(Z) = \sum_{i=1}^n S_{m-1}(\mathrm{Tr}_i\, Z)$, where each $\mathrm{Tr}_i\, Z$ is an operator on $n-1$ factors.

Partial traces do not increase the trace norm. Combining this with the induction hypothesis on $n-1$ factors and Theorem 1 gives

$$mS_m(Z) \leq \tbinom{n-2}{m-1}\big[(n-1)\|Z\|_2^2 + \|Z\|_1^2\big] + n\tbinom{n-2}{m-2}\|Z\|_1^2.$$

To finish, use $(n-1)\binom{n-2}{m-1} = m\binom{n-1}{m}$ and $\binom{n-2}{m-1} + n\binom{n-2}{m-2} = m\binom{n-1}{m-1}$. The second identity follows from $\binom{n-2}{m-1} = \frac{n-m}{m-1}\binom{n-2}{m-2}$.

For the equality case: for $Z = P \otimes |c\rangle\langle c|$, one has $S_m(Z) = \binom{n-1}{m-1}\|P\|_1^2 + \binom{n-1}{m}\|P\|_2^2$. For rank two, $\|C\|_1^2 \leq 2\|C\|_2^2$ turns the bound into (2). □

Proof of Corollaries 4 and 5. For $\rho \geq 0$ we have $\|\rho\|_1 = \mathrm{Tr}\,\rho$; this gives Corollary 4.

For Corollary 5 it suffices, by linearity, to take $\sigma = |\psi\rangle\langle\psi|$. For any vector φ and any set T of factors, the swap trick gives $\langle\varphi|\sigma_T \otimes I|\varphi\rangle = \|\mathrm{Tr}_T(|\psi\rangle\langle\varphi|)\|_2^2$, where σ_T is the marginal on T and Tr_T traces out T. Hence, with $Z = \mathrm{Tr}_{A_0}(|\psi\rangle\langle\varphi|)$,

$$\langle\varphi|\Lambda(\sigma)|\varphi\rangle = \|\psi\|^2\|\varphi\|^2 + (n-1)\|Z\|_2^2 - \sum_i \|\mathrm{Tr}_i\, Z\|_2^2.$$

This is nonnegative by Theorem 1, since $\|Z\|_1 \leq \|\psi\|\|\varphi\|$.

The last statement is $\mathrm{Tr}[\sigma\Lambda(\sigma)] \geq 0$. □

5 Remarks and open problems

Other multipartite forms. The bipartite inequality of Wu and Zou, with $|\,\mathrm{Tr}\,X|^2$ in place of $\|X\|_2^2$, does not extend in the same way. The bound $\sum_i \|\mathrm{Tr}_i\, Z\|_2^2 \leq (n-2)\|Z\|_2^2 + \|Z\|_1^2 + |\,\mathrm{Tr}\,Z|^2$ fails for $n = 3$: a numerical search finds rank-two operators on three qubits that violate it by about 2.5% of $\|Z\|_1^2$.

A sharper rank-two bound. For $n = 2$ the sharp rank-two inequality of [6, 7] reads $\|\mathrm{Tr}_1 C\|_2^2 + \|\mathrm{Tr}_2 C\|_2^2 \leq 2\|C\|_2^2 + \frac{1}{2}|\,\mathrm{Tr}\,C|^2$. Numerically, the multipartite version

$$\sum_{i=1}^n \|\mathrm{Tr}_i\, C\|_2^2 \leq n\|C\|_2^2 + \tfrac{1}{2}|\,\mathrm{Tr}\,C|^2 \qquad (\mathrm{rank}\, C \leq 2)$$

holds in our searches, with equality attained, for $n \leq 6$ $(d = 2)$, $n = 3, \ldots, 6, 8, 10, 12$ $(d = 3)$, $n = 4, 6$ $(d = 4)$ and $n = 3, 5$ $(d = 5)$. For positive C it follows from Corollary 4, since then $(\mathrm{Tr}\,C)^2 \leq 2\,\mathrm{Tr}\,C^2$. It then holds for every normal C of rank two. Write $C = \lambda_1 uu^* + \lambda_2 vv^*$ with orthonormal u, v, put $a = |\lambda_1|$, $b = |\lambda_2|$, $r = \mathrm{Re}(\bar\lambda_1\lambda_2)$ and $C_+ = a\,uu^* + b\,vv^*$. Then $S_1(C) = a^2 S_1(uu^*) + b^2 S_1(vv^*) + 2rc$ with $c = \sum_i \langle \mathrm{Tr}_i\, uu^*, \mathrm{Tr}_i\, vv^*\rangle \geq 0$. If $r \leq 0$, the rank-one bound $S_1(uu^*) \leq n$ gives $S_1(C) \leq n\|C\|_2^2$. If $r > 0$, put $\theta = r/(ab) \leq 1$. Then $S_1(C) = (1-\theta)\,(a^2 S_1(uu^*) + b^2 S_1(vv^*)) + \theta\, S_1(C_+) \leq n(a^2+b^2) + \theta(a+b)^2/2$, and this is at most $n\|C\|_2^2 + \frac{1}{2}|\,\mathrm{Tr}\,C|^2 = n(a^2+b^2) + \frac{1}{2}(a^2+b^2+2r)$.

The bound also holds whenever the ranges of C and C^* together span at most two dimensions, that is, for $C = VMV^*$ with $V : \mathbb{C}^2 \to H$ an isometry and M any 2×2 matrix. Write $C = Z_1 + iZ_2$ with $Z_1 = (C + C^*)/2$ and $Z_2 = (C - C^*)/2i$. Both are Hermitian of rank at most two. For Hermitian X, Y the numbers $\langle \mathrm{Tr}_J X, \mathrm{Tr}_J Y\rangle$ are real, so $\|\mathrm{Tr}_J C\|_2^2 = \|\mathrm{Tr}_J Z_1\|_2^2 + \|\mathrm{Tr}_J Z_2\|_2^2$ for every J. Hence $F(C) = S_1(C) - n\|C\|_2^2 - \frac{1}{2}|\operatorname{Tr} C|^2$ satisfies $F(C) = F(Z_1) + F(Z_2) \le 0$ by the Hermitian case. The same splitting shows that for such C the endpoint inequality (1) reduces to Hermitian operators of rank two. When $\operatorname{ran} C + \operatorname{ran} C^*$ has dimension three or four we do not have a proof; the Hermitian part of C can then have rank four, and F can be positive on it.

Sharp forms with a flatness term. Let $\sigma_1 \ge \sigma_2$ be the singular values of a rank-two C. For two factors the sharp rank-two inequality can be improved to

$$\|\mathrm{Tr}_1 C\|_2^2 + \|\mathrm{Tr}_2 C\|_2^2 \le \|C\|_2^2 + \tfrac{1}{2}\|C\|_1^2 + \tfrac{1}{2}|\operatorname{Tr} C|^2, \qquad \text{i.e.} \qquad q_2(C) \ge \tfrac{1}{4}(\sigma_1 - \sigma_2)^2.$$

Since $\|C\|_1^2 = \|C\|_2^2 + 2\sigma_1\sigma_2$, this is stronger than the bound of [6, 7] by $\frac{1}{2}(\sigma_1 - \sigma_2)^2$. It is an equality for every $C = \rho \otimes |c\rangle\langle c|$ with $\rho \ge 0$ of rank two, not only for flat ρ. It follows from the proof in [6]. There $C = x_1 y_1^* + x_2 y_2^*$ with $\langle x_i, x_j\rangle = \langle y_i, y_j\rangle = g_{ij}$ and $\langle y_i, x_i\rangle = \operatorname{Tr} C/2$. To build it, take a singular value decomposition $C = U\Sigma V^*$, put $X_0 = U\Sigma^{1/2}$ and $Y_0 = V\Sigma^{1/2}$, and replace them by $X_0 Q$ and $Y_0 Q$, where the unitary Q makes both diagonal entries of $Q^* Y_0^* X_0 Q$ equal to $\operatorname{Tr} C/2$. This is possible because the numerical range of a 2×2 matrix contains half its trace. Then $G = (g_{ij}) = Q^*\Sigma Q$, so $g_{11} + g_{22} = \|C\|_1$ and $\|C\|_2^2 = g_{11}^2 + g_{22}^2 + 2|g_{12}|^2$. For this decomposition we have the identity

$$\|C\|_2^2 + \tfrac{1}{2}\|C\|_1^2 + \tfrac{1}{2}|\operatorname{Tr} C|^2 - \sum_{i=1,2}\|\mathrm{Tr}_i C\|_2^2 = \tfrac{1}{2}(g_{11} - g_{22})^2 + 4\sum_{(a,b)}\|\Pi(a\otimes b)\|^2 + \sum_{i=1,2}\|\mathrm{Tr}_i(x_2 x_1^* - y_2 y_1^*)\|_2^2.$$

Here $\Pi = \frac{1}{4}(I - F_1)(I - F_2)$ on two copies of $H_1 \otimes H_2$, F_i swaps factor i between the copies, and (a, b) runs over (x_1, y_1), (x_2, y_2), (x_1, x_2), (y_1, y_2). Every term on the right is nonnegative.

To prove the identity, write $\Phi(Z) = (\mathrm{Tr}_1 Z, \mathrm{Tr}_2 Z)$, so that $\sum_i \|\mathrm{Tr}_i C\|_2^2 = \|\Phi(C)\|^2$. The swap trick gives $\langle \Phi(uv^*), \Phi(st^*)\rangle = \langle u \otimes t | F_1 + F_2 | v \otimes s\rangle$ for all vectors u, v, s, t. Since $F_1 F_2$ exchanges the two copies, this has two consequences:

$$\langle \Phi(uv^*), \Phi(st^*)\rangle = \langle \Phi(tv^*), \Phi(su^*)\rangle, \qquad \|\Phi(xy^*)\|^2 = \|x\|^2\|y\|^2 + |\langle x, y\rangle|^2 - 4\|\Pi(x\otimes y)\|^2,$$

the second because $4\Pi = I - F_1 - F_2 + F_1 F_2$. Now expand $\|\Phi(C)\|^2$. By the first relation the cross term is $2\operatorname{Re}\langle \Phi(y_2 y_1^*), \Phi(x_2 x_1^*)\rangle = \|\Phi(x_2 x_1^*)\|^2 + \|\Phi(y_2 y_1^*)\|^2 - \|\Phi(x_2 x_1^* - y_2 y_1^*)\|^2$. Applying the second relation to the four rank-one terms gives $\|\Phi(C)\|^2 = g_{11}^2 + g_{22}^2 + \frac{1}{2}|\operatorname{Tr} C|^2 + 2g_{11}g_{22} + 2|g_{12}|^2 - 4\sum_{(a,b)}\|\Pi(a\otimes b)\|^2 - \|\Phi(x_2 x_1^* - y_2 y_1^*)\|^2$, which is the identity. For rank r the same steps, starting from [6, Eq. (65)], give $\|C\|_2^2 + (1 - \frac{1}{r})\|C\|_1^2 + \frac{1}{r}|\operatorname{Tr} C|^2$ on the right, which is again an equality for $\rho \otimes |c\rangle\langle c|$ with $\rho \ge 0$ of rank r.

For n factors the analogues would be

$$q_n(C) \ge 2^{-n}(\sigma_1 - \sigma_2)^2, \qquad S_1(C) \le (n-1)\|C\|_2^2 + \tfrac{1}{2}\big(\|C\|_1^2 + |\operatorname{Tr} C|^2\big) \qquad (\operatorname{rank} C \le 2).$$

Both are equalities for $\rho \otimes (cc^*)^{\otimes(n-1)}$ with $\rho \ge 0$ of rank two. In our searches the first holds for $3 \le n \le 10$ at $d = 3$, and the second for $n = 3, 4, 6, 8, 10$ at $d = 3$, with spot checks at $d = 2, 4, 5, 6$. For rank one the first is [5, Prop. 8]. The second adds $\frac{1}{2}(\sigma_1 - \sigma_2)^2$ to the slack of the sharp rank-two bound above, and it holds for normal C: in the argument for normal C above, Theorem 1 applied to C_+ gives $S_1(C) \le n\|C\|_2^2 + 2\max(r, 0)$, while the right-hand side equals $n\|C\|_2^2 + ab + r$ and $|r| \le ab$. We could not prove either statement in general for $n \ge 3$: the block induction behind Theorem 1

fails for the second already at $n = 3$, whichever factor is split off. Tomar's Conjecture 7.1 [10] strengthens (1) in a different direction, with a margin given by the third eigenvalue of the one-copy reductions of the co-range. For two copies we prove a margin of that kind below.

The second statement has a natural version for rank at most r:

$$S_1(C) \le (n-1)\|C\|_2^2 + \big(1 - \tfrac{1}{r}\big)\|C\|_1^2 + \tfrac{1}{r}|\operatorname{Tr} C|^2.$$

It is an equality for $\rho \otimes (cc^*)^{\otimes(n-1)}$ with $\rho \ge 0$ of rank r, and as $r \to \infty$ it becomes Theorem 1. For $n = 2$ it is the bound proved above. For $r = 1$ it holds for every n: it then reads $S_1(xy^*) \le (n-1)\|x\|^2\|y\|^2 + |\langle y, x\rangle|^2$, that is $\sum_i \rho_i \otimes I \le (n-1)(\operatorname{Tr}\rho)\, I + \rho$ for $\rho = xx^*$, where ρ_i is the marginal on factor i and I acts on the other factors. For $n = 2$ this is [6, Lemma 6.4]. For larger n, apply the statement for $n-1$ factors to the marginal of ρ on the first $n-1$ factors, add $\rho_n \otimes I$, and use the case $n = 2$ for the cut between the first $n-1$ factors and the last. In our searches the rank-r statement also holds for $r = 3, 4$ at $n = 3, 4$.

A thickness margin for two copies. For a two-dimensional subspace $V \subseteq H_1 \otimes H_2$ and $l = 1, 2$, let $\tau_l(V)$ be the sum of all but the two largest eigenvalues of the reduced operator of the projector P_V on factor l. It is at least the third eigenvalue $\lambda_3^{(l)}(V)$ used by Tomar, and it vanishes exactly when the local support of V at factor l has dimension at most two.

Proposition 7. *Let C have rank at most two on $H_1 \otimes H_2$, and let V be a two-dimensional subspace that contains the range or the co-range of C. Then*

$$q_2(C) \ge \frac{\min\{\tau_1(V), \tau_2(V)\}}{48 - 8\sqrt{2}}\, \|C\|_2^2.$$

In the notation of [10] this gives $m_2(V) \ge \frac{1}{37} \min_l \lambda_3^{(l)}(V)$ for every plane V. That is the case $k = 2$ of Tomar's Conjecture 7.1 with a smaller constant: he conjectures $\frac{1}{4}$, and in our searches the best constant at $d = 3$ is about 0.29, in line with Tomar's numerics. For two copies no linear bound of this kind was known [10, Sec. 8]. In our searches the margin is even linear in the larger tail $\max_l \tau_l(V)$, with infimum $(3 - \sqrt{5})/4 \approx 0.19$ attained at $V = \operatorname{span}\{|00\rangle, (|01\rangle + |12\rangle)/\sqrt{2}\}$, where $m_2(V) = (3 - \sqrt{5})/8$; our argument does not give this. The proof uses two lemmas.

Lemma 8. *Let $\mathcal{A} \ge 0$ be an operator on $\mathbb{C}^2 \otimes K$ with blocks $\mathcal{A}_{jk}$ on K, and suppose that $\operatorname{Tr}_K \mathcal{A} = I_2$, that is, $\operatorname{Tr}\mathcal{A}_{00} = \operatorname{Tr}\mathcal{A}_{11} = 1$ and $\operatorname{Tr}\mathcal{A}_{01} = 0$.*

(a) $\lambda_{\max}(\mathcal{A}) \le 1 + \frac{1}{\sqrt{2}}\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2$.

(b) Let ψ be a unit vector with $\langle\psi|\mathcal{A}|\psi\rangle = 1 + \eta$ and $\eta \ge 0$, let E be the projector onto the support of the reduced operator of $\psi\psi^*$ on K, and put $\varepsilon = \operatorname{Tr}\big((I \otimes (I - E))\mathcal{A}\big)$. Then $\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \ge 2\eta(\eta + \varepsilon)$.

In particular, if $\lambda = \lambda_{\max}(\mathcal{A}) \ge 1$ and τ is the sum of all but the two largest eigenvalues of $\mathcal{A}_{00} + \mathcal{A}_{11}$, then $\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \ge 2(\lambda - 1)(\lambda - 1 + \tau)$.

The last statement follows from (b) with ψ a top eigenvector, because E has rank at most two, so that $\varepsilon = \operatorname{Tr}\big((I - E)(\mathcal{A}_{00} + \mathcal{A}_{11})\big) \ge \tau$. Part (a) is sharp: equality holds for $\mathcal{A} = \lambda\,\varphi\varphi^* + (2 - \lambda)\,\varphi'\varphi'^*$ with $1 \le \lambda \le 2$, $\varphi = (|0\rangle a + |1\rangle b)/\sqrt{2}$, $\varphi' = (|0\rangle b - |1\rangle a)/\sqrt{2}$ and $a \perp b$ unit vectors.

Proof. Let ψ be a unit vector, with Schmidt decomposition $\psi = \sqrt{p}\, q_1 \otimes a + \sqrt{1 - p}\, q_2 \otimes b$, where $a, b \in K$ are orthonormal (enlarging K if necessary). Conjugating $\mathcal{A}$ by a diagonal unitary U on $\mathbb{C}^2$, and replacing ψ by $(U \otimes I)\psi$, changes neither the hypotheses, nor the spectrum, nor $\mathcal{A}_{00} - \mathcal{A}_{11}$,

nor $\langle\psi|\mathcal{A}|\psi\rangle$, E and ε in (b). So we may assume that $q_1 = (c, s)$ and, after a phase change of b, $q_2 = (-s, c)$, where $c = \cos\frac{\theta}{2}$, $s = \sin\frac{\theta}{2}$ and $0 \le \theta \le \pi$. Put $r = \sqrt{p(1-p)}$ and

$$Z = \tfrac{1}{2}I + (p - \tfrac{1}{2})\sin\theta\,\sigma_x, \qquad Y = \tfrac{1}{\sqrt{2}}\big[\cos\theta\,(aa^* - bb^*) - 2r\sin\theta\,(ab^* + ba^*)\big].$$

Then $\operatorname{Tr} Z = 1$, $Z \ge 0$ and $\|Y\|_2^2 = \cos^2\theta + 4r^2\sin^2\theta \le 1$. We claim that $\psi\psi^* \le Z \otimes I + \frac{1}{\sqrt{2}}\sigma_z \otimes Y$. Both sides commute with $I \otimes E$, where E is the projector onto $\operatorname{span}\{a, b\}$, so we may check the two blocks separately. On $\mathbb{C}^2 \otimes \{a, b\}^\perp$ the left side vanishes and the right side is $Z \otimes I \ge 0$. On $\mathbb{C}^2 \otimes \operatorname{span}\{a, b\}$, in the basis $|0a\rangle, |0b\rangle, |1a\rangle, |1b\rangle$, the right side is

$$P = \begin{pmatrix} c^2 & -2rcs & \kappa & 0 \\ -2rcs & s^2 & 0 & \kappa \\ \kappa & 0 & s^2 & 2rcs \\ 0 & \kappa & 2rcs & c^2 \end{pmatrix}, \qquad \kappa = (2p-1)cs,$$

and $\psi = (\sqrt{p}\,c,\ -\sqrt{1-p}\,s,\ \sqrt{p}\,s,\ \sqrt{1-p}\,c)$. Using $c^2 + s^2 = 1$ and $\kappa^2 + 4r^2c^2s^2 = c^2s^2$, one checks that $P^2 = P$ and $P\psi = \psi$. So P is a projector whose range contains ψ, and $P - \psi\psi^* \ge 0$. Hence

$$\langle\psi|\mathcal{A}|\psi\rangle \le \operatorname{Tr}\big(Z \operatorname{Tr}_K \mathcal{A}\big) + \tfrac{1}{\sqrt{2}}\operatorname{Tr}\big(Y(\mathcal{A}_{00} - \mathcal{A}_{11})\big) \le 1 + \tfrac{1}{\sqrt{2}}\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2,$$

which is (a) when ψ is an eigenvector for $\lambda_{\max}(\mathcal{A})$.

For (b) we may assume $\eta > 0$. Then $0 < p < 1$, since $\langle\psi|\mathcal{A}|\psi\rangle \le 1$ when ψ is a product vector, and E is the projector onto $\operatorname{span}\{a, b\}$. Put $u = |p - \frac{1}{2}|\sin\theta$ and $\ell = \frac{1}{2} - u$, the smallest eigenvalue of Z; then $u < \frac{1}{2}$, so $\ell > 0$. Since $4r^2 = 1 - (2p-1)^2$, we have $\|Y\|_2^2 = \cos^2\theta + 4r^2\sin^2\theta = 1 - 4u^2 = 4\ell(1-\ell)$. Let $\gamma = 2\eta/\ell$. By the two blocks above, $Z \otimes I + \frac{1}{\sqrt{2}}\sigma_z \otimes Y - \psi\psi^*$ equals $P - \psi\psi^* \ge 0$ on $\mathbb{C}^2 \otimes \operatorname{span}\{a, b\}$ and $Z \otimes I \ge \ell I$ on $\mathbb{C}^2 \otimes \{a, b\}^\perp$. Hence $\gamma\big(Z \otimes I + \frac{1}{\sqrt{2}}\sigma_z \otimes Y - \psi\psi^*\big) - 2\eta\, I \otimes (I - E) \ge 0$. Pairing this with $\mathcal{A}$ and using $\operatorname{Tr}(Z \operatorname{Tr}_K \mathcal{A}) = 1$ gives $\frac{1}{\sqrt{2}}\operatorname{Tr}\big(Y(\mathcal{A}_{00} - \mathcal{A}_{11})\big) \ge \eta + 2\eta\varepsilon/\gamma = \eta + \ell\varepsilon$. By the Cauchy–Schwarz inequality,

$$\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \ge \frac{2(\eta + \ell\varepsilon)^2}{\|Y\|_2^2} = \frac{(\eta + \ell\varepsilon)^2}{2\ell(1-\ell)} \ge 2\eta(\eta + \varepsilon),$$

where the last step is $(\eta + \ell\varepsilon)^2 - 4\ell(1-\ell)\eta(\eta + \varepsilon) = \big((1-2\ell)\eta - \ell\varepsilon\big)^2$. □

Lemma 9. *Let w_1, w_2 be an orthonormal basis of a two-dimensional subspace $V \subseteq H_1 \otimes H_2$, and let Π be as in the identity above. Then*

$$4\|\Pi(w_1 \otimes w_2)\|^2 \ge \tfrac{1}{2}\min\{\tau_1(V), \tau_2(V)\}.$$

The constant $\frac{1}{2}$ is best possible. If $\dim H_1 = \dim H_2 = d \ge 3$ and $\sqrt{d}\,w_1$, $\sqrt{d}\,w_2$ are unitary $d \times d$ matrices that are orthogonal in the Hilbert–Schmidt inner product, then the reduced operators A_j, B_j in the proof below all equal I/d, and both sides equal $1 - 2/d$.

Proof. For vectors x_1, x_2 let A_j and B_j be the reduced operators of $x_j x_j^*$ on H_1 and H_2. Expanding $\Pi = \frac{1}{4}(I - F_1 - F_2 + F_1F_2)$ gives

$$4\|\Pi(x_1 \otimes x_2)\|^2 = \|x_1\|^2\|x_2\|^2 + |\langle x_1, x_2\rangle|^2 - \operatorname{Tr} A_1A_2 - \operatorname{Tr} B_1B_2. \tag{7}$$

Now take $x_j = w_j$ and write $\Delta = 4\|\Pi(w_1 \otimes w_2)\|^2$. Then $\rho_1 = A_1 + A_2$ and $\rho_2 = B_1 + B_2$ are the reductions of P_V. Let $\mu_1 \ge \mu_2 \ge \dots$ be the eigenvalues of ρ_2. They sum to two, and $\tau_2 = \sum_{i \ge 3}\mu_i$.

(i) By (7), $\Delta = 1 - \operatorname{Tr} A_1A_2 - \operatorname{Tr} B_1B_2$. Since $\operatorname{Tr} B_j^2 = \operatorname{Tr} A_j^2$, this equals $1 - \frac{1}{2}\operatorname{Tr}\rho_2^2 + \frac{1}{2}\|A_1 - A_2\|_2^2$.

(ii) Since $\mu_i \le \mu_2$ for $i \ge 3$ and $\mu_1 + \mu_2 + \tau_2 = 2$,

$$2 - \operatorname{Tr}\rho_2^2 - \tau_2 = \mu_1(1-\mu_1) + \mu_2(1-\mu_2) - \sum_{i\ge 3}\mu_i^2 \ge \mu_1(1-\mu_1) + \mu_2(\mu_1 - 1) = (1-\mu_1)(\mu_1 - \mu_2).$$

(iii) The operator $\mathcal{A}$ on $\mathbb{C}^2 \otimes H_1$ with blocks $\mathcal{A}_{jk} = \operatorname{Tr}_2(w_{j+1}w_{k+1}^*)$, $j,k \in \{0,1\}$, is the reduced operator of the vector $|0\rangle \otimes w_1 + |1\rangle \otimes w_2$. So its nonzero eigenvalues are those of ρ_2. Moreover $\operatorname{Tr}_{H_1}\mathcal{A} = I_2$, $\mathcal{A}_{00} - \mathcal{A}_{11} = A_1 - A_2$ and $\mathcal{A}_{00} + \mathcal{A}_{11} = \rho_1$. By the last statement of Lemma 8, $\|A_1 - A_2\|_2^2 \ge 2(\mu_1 - 1)(\mu_1 - 1 + \tau_1)$ whenever $\mu_1 \ge 1$.

If $\mu_1 \le 1$, then (i) and (ii) give $\Delta \ge \frac{1}{2}\tau_2$. If $\mu_1 > 1$, then (i)–(iii) and $\mu_1 - \mu_2 = 2(\mu_1 - 1) + \tau_2$ give

$$\Delta \ge \tfrac{1}{2}\tau_2 - \tfrac{1}{2}(\mu_1 - 1)(\mu_1 - \mu_2) + (\mu_1 - 1)^2 + (\mu_1 - 1)\tau_1 = \tfrac{1}{2}(2 - \mu_1)\,\tau_2 + (\mu_1 - 1)\,\tau_1.$$

The right side is at least $\frac{1}{2}\mu_1 \min\{\tau_1, \tau_2\} \ge \frac{1}{2}\min\{\tau_1, \tau_2\}$. □

Proof of Proposition 7. Since $q_2(C^*) = q_2(C)$, we may assume that V contains the range of C. Write $m = \min\{\tau_1(V), \tau_2(V)\}$; note $m \le 2$. If $\operatorname{rank} C \le 1$, then $q_2(C) \ge \frac{1}{4}\|C\|_2^2$ by [5, Prop. 8], which is enough. So let $\operatorname{rank} C = 2$, so that V is the range of C, and take the decomposition $C = x_1y_1^* + x_2y_2^*$ used for the identity above. Its Gram matrix $G = (g_{ij})$ has eigenvalues $\sigma_1 \ge \sigma_2 > 0$, and x_1, x_2 span V. Put $s = \frac{1}{2}(\sigma_1 + \sigma_2)$ and $\delta = \frac{1}{2}(\sigma_1 - \sigma_2)$. Since every term on the right of the identity is nonnegative, and $2q_2(C)$ is its left side plus $\frac{1}{2}(\sigma_1 - \sigma_2)^2$,

$$q_2(C) \ge \delta^2 + 2\|\Pi(x_1 \otimes x_2)\|^2.$$

Write $G = Q^* \operatorname{diag}(\sigma_1, \sigma_2) Q$ with Q unitary. Then $x_j = \sum_k \sqrt{\sigma_k}\, Q_{kj} v_k$ for an orthonormal basis v_1, v_2 of V. Let $\hat{w}_j = \sum_k Q_{kj} v_k$, which is another orthonormal basis of V. Since $\Pi(u \otimes v) = \Pi(v \otimes u)$,

$$\begin{aligned}\Pi(x_1 \otimes x_2) - s\,\Pi(\hat{w}_1 \otimes \hat{w}_2) = {}& \delta Q_{11}Q_{12}\Pi(v_1 \otimes v_1) - \delta Q_{21}Q_{22}\Pi(v_2 \otimes v_2) \\ & + (\sqrt{\sigma_1\sigma_2} - s)(Q_{11}Q_{22} + Q_{21}Q_{12})\Pi(v_1 \otimes v_2).\end{aligned}$$

By (7), $\|\Pi(v_k \otimes v_k)\|^2 \le \frac{1}{2}$ and $\|\Pi(v_1 \otimes v_2)\|^2 \le \frac{1}{4}$. Moreover $|Q_{11}Q_{12}|, |Q_{21}Q_{22}| \le \frac{1}{2}$, $|Q_{11}Q_{22} + Q_{21}Q_{12}| \le 1$ and $0 \le s - \sqrt{\sigma_1\sigma_2} \le \delta$. So the difference has norm at most $\beta\delta$ with $\beta = \frac{1}{\sqrt{2}} + \frac{1}{2}$, and $4\beta^2 = 3 + 2\sqrt{2}$. By Lemma 9, $\|\Pi(\hat{w}_1 \otimes \hat{w}_2)\| \ge \sqrt{\nu}$ with $\nu = m/8 \le \frac{1}{4}$. Hence, with $\zeta = \delta/s \in [0,1]$,

$$q_2(C) \ge \delta^2 + 2\big(s\sqrt{\nu} - \beta\delta\big)_+^2 = s^2 g(\zeta), \qquad g(\zeta) = \zeta^2 + 2\big(\sqrt{\nu} - \beta\zeta\big)_+^2,$$

while $\|C\|_2^2 = \sigma_1^2 + \sigma_2^2 = 2s^2(1 + \zeta^2)$. If $\beta\zeta \ge \sqrt{\nu}$, then $g(\zeta)/(2(1+\zeta^2)) \ge \zeta^2/(2(1+\zeta^2)) \ge \nu/(2\beta^2 + 2\nu)$, since $\zeta^2/(1+\zeta^2)$ increases with ζ. If $\beta\zeta < \sqrt{\nu}$, then $g(\zeta) \ge \min_{z\in\mathbb{R}}\{z^2 + 2(\sqrt{\nu} - \beta z)^2\} = 2\nu/(1 + 2\beta^2)$ and $1 + \zeta^2 < 1 + \nu/\beta^2$. Since $2\beta^2 + 2\nu \le (1 + 2\beta^2)(1 + \nu/\beta^2)$, in both cases

$$\frac{q_2(C)}{\|C\|_2^2} \ge \frac{\nu}{(1 + 2\beta^2)(1 + \nu/\beta^2)} \ge \frac{\nu}{(\frac{5}{2} + \sqrt{2})(4 - 2\sqrt{2})} = \frac{\nu}{6 - \sqrt{2}} = \frac{m}{48 - 8\sqrt{2}},$$

where we used $\nu/\beta^2 \le 1/(4\beta^2) = 3 - 2\sqrt{2}$. □

Range and co-range in one plane. For these operators, in particular for normal ones, the argument gives Tomar's inequality with the best possible constant. Tomar proved this constant for every operator whose range lies in a plane spanned by two vectors with common Schmidt bases [10, Thm. 7.2]; here the plane is arbitrary, but the co-range must lie in it too. If the range and co-range of C lie in a plane V, then

$$q_2(C) \geq \big(1 - \tfrac{1}{\sqrt{2}}\big) \min_l \lambda_3^{(l)}(V)\, \|C\|_2^2,$$

and $1 - 1/\sqrt{2} > \frac{1}{4}$. First, $q_2(C) = q_2(\operatorname{Re} C) + q_2(\operatorname{Im} C)$ and $\|C\|_2^2 = \|\operatorname{Re} C\|_2^2 + \|\operatorname{Im} C\|_2^2$ for every C, because partial traces of Hermitian operators are Hermitian and $\operatorname{Tr}(PQ)$ is real for Hermitian P, Q (Wu and Zou use the same splitting for three copies [8, Cor. 6.2]); so it suffices to treat Hermitian C with range in V, and rank one is immediate. For $C = \omega_1 v_1 v_1^* + \omega_2 v_2 v_2^*$ with v_1, v_2 orthonormal, $\sigma_j = |\omega_j|$, $\sigma_1 \geq \sigma_2 > 0$ and $\omega_1 \bar{\omega}_2 = \sigma_1 \sigma_2 e^{i\varphi}$, a direct computation using (7) gives

$$q_2(C) = \sigma_1^2\big(\tfrac{1}{4} + E_1\big) + \sigma_2^2\big(\tfrac{1}{4} + E_2\big) + \sigma_1\sigma_2 \cos\varphi\, \big(\Delta - \tfrac{1}{2}\big),$$

where $E_j = 1 - \operatorname{Tr} \rho_{v_j}^2$ for the reduced operator ρ_{v_j} of $v_j v_j^*$ and $\Delta = 4\|\Pi(v_1 \otimes v_2)\|^2 \in [0,1]$. Since $|\cos\varphi\,(\Delta - \frac{1}{2})| \leq \frac{1}{2}$, for any κ and μ we get $q_2(C) - \kappa\mu\|C\|_2^2 \geq k_1\sigma_1^2 + k_2\sigma_2^2 - \frac{1}{2}\sigma_1\sigma_2$ with $k_j = \frac{1}{4} + E_j - \kappa\mu$, which is nonnegative once $k_1, k_2 > 0$ and $k_1 k_2 \geq \frac{1}{16}$. Put $e_j = \min\{\frac{1}{2}, E_j\}$. Weyl's inequality $\lambda_3(P+Q) \leq \lambda_2(P) + \lambda_2(Q)$ gives $\mu := \min_l \lambda_3^{(l)}(V) \leq s := e_1 + e_2$, and for $\kappa = 1 - 1/\sqrt{2}$ the identity $(\frac{1}{4} + e_1 - \kappa s)(\frac{1}{4} + e_2 - \kappa s) - \frac{1}{16} = \frac{\sqrt{2}-1}{4} s(1-2s) + e_1 e_2$ shows $k_1 k_2 \geq \frac{1}{16}$ (for $s > \frac{1}{2}$ use $e_1 e_2 \geq \frac{1}{2}(s - \frac{1}{2})$). Tomar's example $C = |00\rangle\langle 00| - (\sqrt{2}-1) v_2 v_2^*$ with $v_2 = (|11\rangle + |22\rangle)/\sqrt{2}$ attains the constant. In the same way Ky Fan's maximum principle gives $\max_l \tau_l(V) \leq (1-p_1) + (1-p_2) \leq E_1 + E_2 =: S$ for the largest eigenvalues p_j of ρ_{v_j}, and for $\kappa = (3 - \sqrt{5})/4$, where $\frac{1}{4} - \frac{\kappa}{2} = \kappa(1-\kappa)$, the identity $(\frac{1}{4} + E_1 - \kappa S)(\frac{1}{4} + E_2 - \kappa S) - \frac{1}{16} = \kappa(1-\kappa) S(1-S) + E_1 E_2$ (with $E_1 E_2 \geq S - 1$ when $S > 1$) gives $q_2(C) \geq \frac{3-\sqrt{5}}{4} \max_l \tau_l(V) \|C\|_2^2$ on the same class; this is the constant we found numerically for general C. For $C \geq 0$, using $\Delta \geq \frac{1}{2} \min_l \tau_l(V)$ from Lemma 9, a similar computation gives $q_2(C) \geq \frac{1}{4} \min_l \tau_l(V) \|C\|_2^2$, with the tails in place of the third eigenvalues. For indefinite C the third eigenvalues in the first bound cannot be replaced by the tails: for $v_1 = |00\rangle$, $v_2 = \sum_{i \geq 1} |ii\rangle/\sqrt{d-1}$ and $C = v_1 v_1^* - t v_2 v_2^*$ the smallest ratio $q_2(C)/(\min_l \tau_l \|C\|_2^2)$ is $\frac{1}{4}$ at $d = 4$ and $(5 - \sqrt{13})/6 \approx 0.232$ at $d = 5$, and it tends to $(3 - \sqrt{5})/4$ as $d \to \infty$.

A graded form of the multiplicativity conjecture. For $0 \leq k \leq n$ put

$$A_k(C) = \sum_{m=0}^{k} \Big(-\tfrac{1}{2}\Big)^m \tbinom{n-m}{k-m} S_m(C), \qquad S_0(C) = \|C\|_2^2.$$

Equivalently, $A_k(C) = \sum_{|S|=k} q^{(S)}(C)$, where $q^{(S)}$ is the form (1) for the copies in S, the other copies being left untraced. With $u = 1 - 2t$ one checks the identity

$$\sum_J (-t)^{|J|} \|\operatorname{Tr}_J C\|_2^2 - (1-2t)(1-t)^{n-1}\|C\|_2^2 = \sum_{k=0}^{n} u^{n-k}(1-u)^k \Big[A_k(C) - \tbinom{n-1}{k} 2^{-k} \|C\|_2^2\Big].$$

This is the expansion that Tomar uses to show that the endpoint inequality propagates to smaller t [9, Lemma 2.3(b)]; there every bracket is replaced by $A_k(C) \geq 0$, which gives the weaker bound $(1-2t)^n \|C\|_2^2$. The Bernstein weights $u^{n-k}(1-u)^k$ are nonnegative for $0 \leq t \leq 1/2$. So the conjectured identity of the introduction holds on the whole interval as soon as

$$A_k(C) \geq \tbinom{n-1}{k} 2^{-k} \|C\|_2^2 \qquad (k = 1, \ldots, n,\ \operatorname{rank} C \leq 2).$$

For $k = 1$ this is the case $m = 1$ of (2), proved here, and for $k = n$ it is (1). For $n = 2$ both levels are known, the second being the two-copy theorem, so the identity holds for two copies at every $t \in [0, 1/2]$. For $n = 3$ the middle level, $11S_0 - 4S_1 + S_2 \geq 0$, also follows from known results. One checks

$$11S_0 - 4S_1 + S_2 = \tfrac{5}{6} \cdot 8q_3 + \tfrac{2}{3} \sum_{i=1}^{3} \Big[2S_0 + \tfrac{1}{2} S_3 - \|\mathrm{Tr}_i\, C\|_2^2 - \|\mathrm{Tr}_{jk}\, C\|_2^2 \Big] + \tfrac{1}{6} \big(2S_0 - S_3 \big),$$

where $\{j, k\}$ is the complement of i and $S_3 = |\operatorname{Tr} C|^2$. The bracket is the slack of the two-copy theorem [6, 7] for the bipartition of the three factors into $\{i\}$ and $\{j, k\}$, and $|\operatorname{Tr} C|^2 \leq 2\|C\|_2^2$ for rank two. So, given the three-copy theorem [9], all three levels hold for $n = 3$, and the conjectured identity of the introduction holds for three copies at every $t \in [0, 1/2]$. For $n \geq 4$ no such linear combination exists: a linear program over the two- and three-copy theorems applied to all groupings of the copies (with the other copies as spectators), the bounds (2), Theorem 1 and the bound with a flatness term above finds none for any level $2 \leq k \leq n$ at $n = 4, 5$. In our numerics every level holds, with equality at $C = P \otimes |c\rangle\langle c|^{\otimes(n-1)}$ for a rank-two projector P on one factor, for all $n \leq 6$ at $d = 3$, with spot checks at $d = 2$ and $d = 4$. Summing the two-copy inequality over all pairs of copies gives only $A_2(C) \geq 0$, which misses the required constant.

Below the PPT point. For $0 \leq t \leq 1/d$ the form $\sum_J (-t)^{|J|} \|\mathrm{Tr}_J\, Z\|_2^2$ is positive semidefinite in Z, and there a rank-free version of the conjectured identity seems to hold: for every operator Z on $(\mathbb{C}^d)^{\otimes n}$,

$$\sum_J (-t)^{|J|} \|\mathrm{Tr}_J\, Z\|_2^2 \geq (1 - t)^{n-1} \big(\|Z\|_2^2 - t \|Z\|_1^2 \big) \qquad (0 \leq t \leq 1/d).$$

Its derivative at $t = 0$ is Theorem 1, and it is an equality for $P \otimes |c\rangle\langle c|^{\otimes(n-1)}$ with $P \geq 0$. For rank two and $d = 3$ it would give the conjectured identity of the introduction on $[0, 1/3]$, for every n. At $t = 1/d$ it says that the part of Z that is traceless on every factor has squared norm at least $(1 - 1/d)^{n-1}(\|Z\|_2^2 - \|Z\|_1^2/d)$. In our searches it holds for $(d, n) = (2, 2), (2, 3), (2, 4)$ at $t = 1/2$, for $(3, 2)$ and $(3, 3)$ at several $t \leq 1/3$, and for $(4, 2)$ at $t = 1/5, 1/4$. It fails as soon as $t > 1/d$: for $d = 3$ and $n = 2$ already at $t = 0.34$, with Z of rank three. The one-site induction behind Theorem 1 reduces it to a block inequality, which holds at $t = 0$ (it is Lemma 6) but fails at $t = 1/d$ in a fixed block basis.

Other Tsallis indices. The bound $\sum_i \mathrm{Tr}[(\mathrm{Tr}_i\, \rho)^q] \leq (n - 1) \operatorname{Tr} \rho^q + (\operatorname{Tr} \rho)^q$ also holds in our numerics for $q = 3/2$ and $q = 3$, at every rank. A Han inequality for all Tsallis entropies with $q > 1$ would therefore be natural. For $n = 2$ it is Audenaert's theorem [14]. Our argument is specific to $q = 2$.

6 What the method does not give

The endpoint inequality (1) needs upper bounds on the odd moments S_m and lower bounds on the even ones. Corollary 3 supplies only the first kind. This is why [12] needed a further unproved assumption to reach undistillability.

The trace-norm mechanism behind Theorem 1 also does not extend directly. We tested three natural rank-free forms of (1) numerically. All of them fail already at $d = 3$:

- $\mathrm{Tr}[(Z \otimes Z^*)(2(-1)^N - (-3)^N)] \leq \|Z\|_1^2$, where N counts the locally antisymmetric factors of two copies. It fails for $n = 2$.
- $q_n(X) \geq 0$ for all X with $\|X\|_1^2 \leq 2\|X\|_2^2$. It fails for $n = 2$.

- $q_n(\rho) \geq 0$ for all density matrices with $\operatorname{Tr}\rho^2 \geq 1/2$. It fails for $n = 3$.

So the rank-two constraint has to be used in an essential way at the endpoint. At first order it does not: the bound of Corollary 2 only sees the rank through $\|C\|_1^2 \leq r\|C\|_2^2$.

The same contrast appears for positive maps. Inequality (1) for all rank-two C says that $\mathrm{id}_2 \otimes \psi^{\otimes n}$ is positive, where $\psi(X) = \operatorname{Tr}(X)I - X/2$. Corollary 5 says that the first-order part of this map stays positive with an ancilla of any dimension.

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